

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Register Number	192365013

MODEL PRACTICAL EXAMINATION – SET 2

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Search for Cabs, From, To, Rental, Out Station, Package Type, Hours/Days, etc.
2. Create a simple cloud software application for Hospital Information System for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS. Include the fields of Patient Name, Age, Address, Phone, Email, Attender Name, Attender Phone, Doctor Name, Diseases, etc.
3. Change the hardware compatibility of a VM (either by clone/create a new one) which is already created and configured, and run simple UNIX commands.
4. Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Answer

1. **Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Search for Cabs, From, To, Rental, Out Station, Package Type, Hours/Days, etc.**

AIM: To Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as Car Booking Reservation System

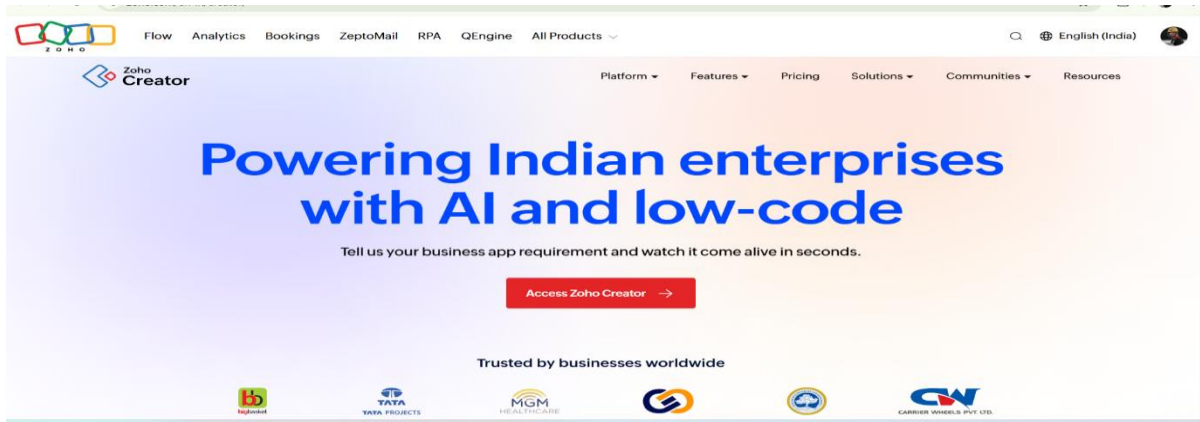
step 5: Created new application Car Booking Reservation System.

step 6: Select one form

step 7: The software has been created

Implementation:

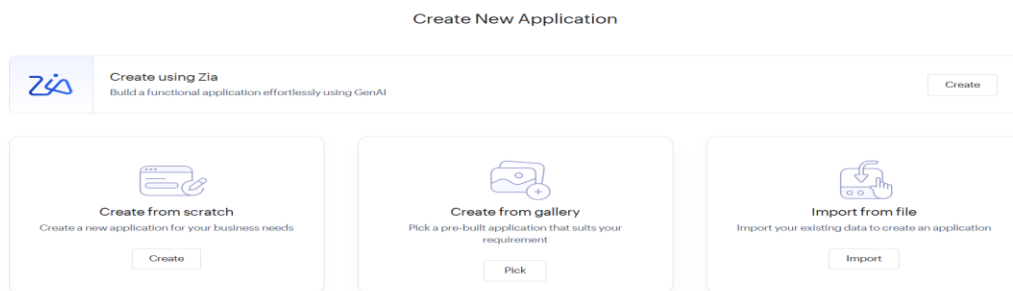
STEP1: GOTO ZOHO.COM



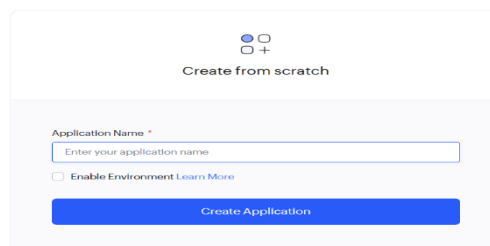
STEP 2: LOGINTOTHE ZOHO.COM



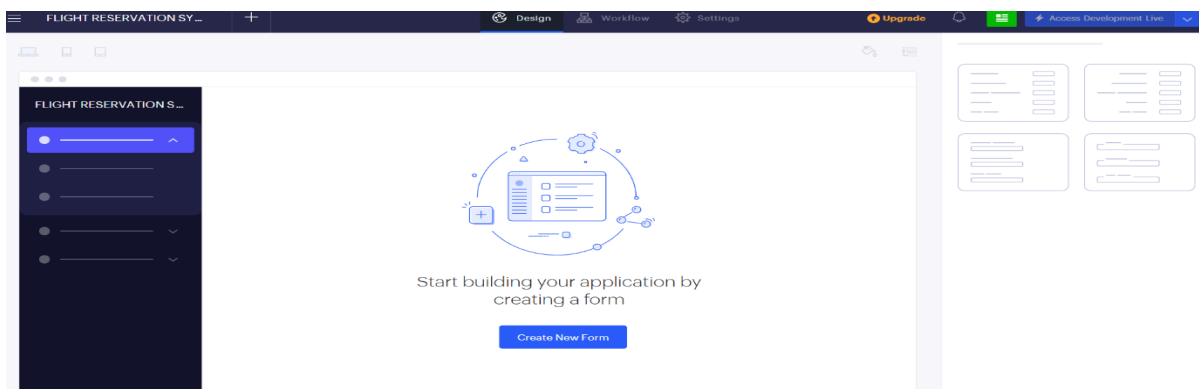
STEP 3: SELECT ONE APPLICATION



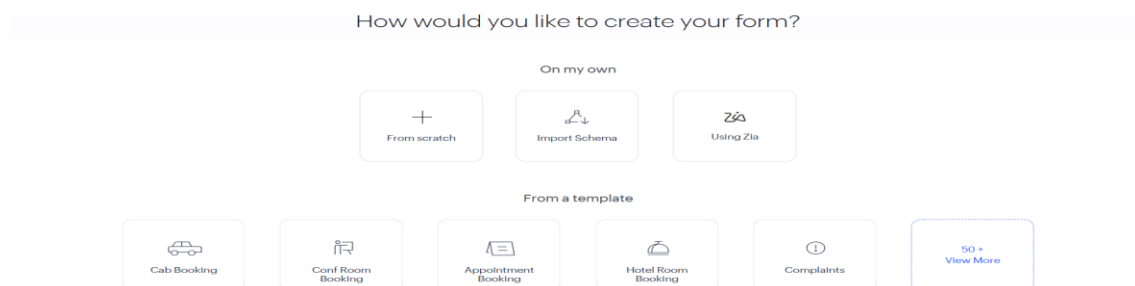
STEP 4: ENTER APPLICATION NAME



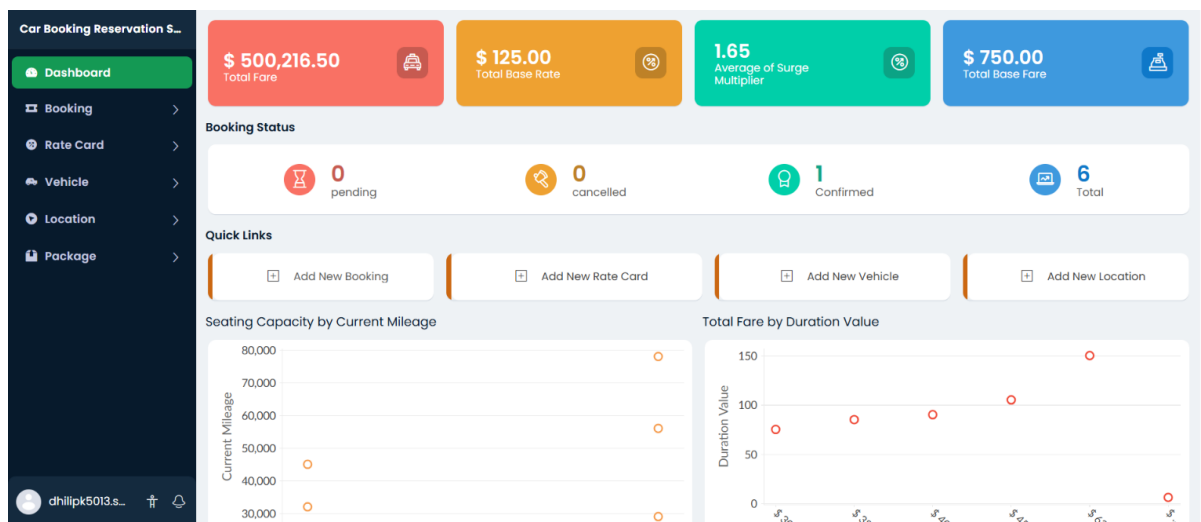
STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HAS BEEN CREATED.



2. **Create a simple cloud software application for Hospital Information System for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS. Include the fields of Patient Name, Age, Address, Phone, Email, Attender Name, Attender Phone, Doctor Name, Diseases, etc.**

AIM:

To create a simple cloud software application for Hospital Information System for SIMATS Hospital using Zoho Creator as a Cloud Service Provider to demonstrate the Software as a Service (SaaS) model.

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open Zoho Creator from the Zoho applications.

Step 4: Click Create New Application.

Step 5: Enter the application name as Hospital Information System.

Step 6: Create a new application.

Step 7: Create a new form named Patient Information.

Step 9: Configure the form to store and manage patient information securely.

Step 10: Create reports to display patient details, doctor information, and treatment records.

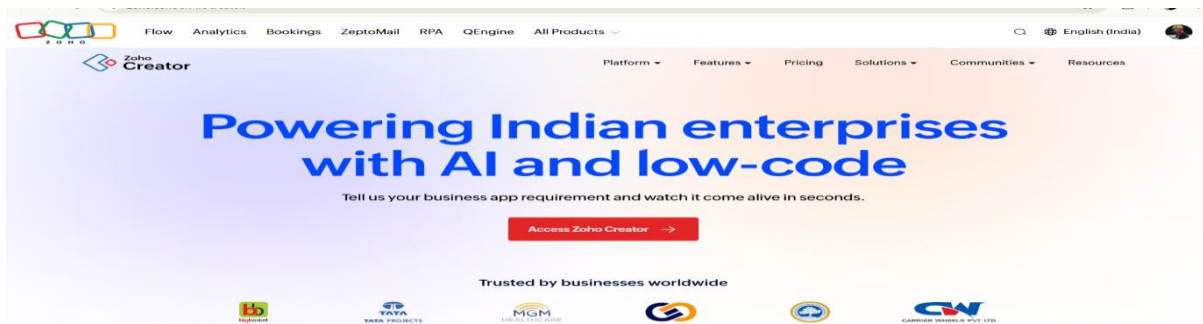
Step 11: Save the application and test it by entering sample patient details.

Step 12: Verify that all patient information is stored correctly and the reports display the required details.

Step 13: The Hospital Information System software has been successfully created and deployed on the Zoho cloud platform.

IMPLEMENTATION:

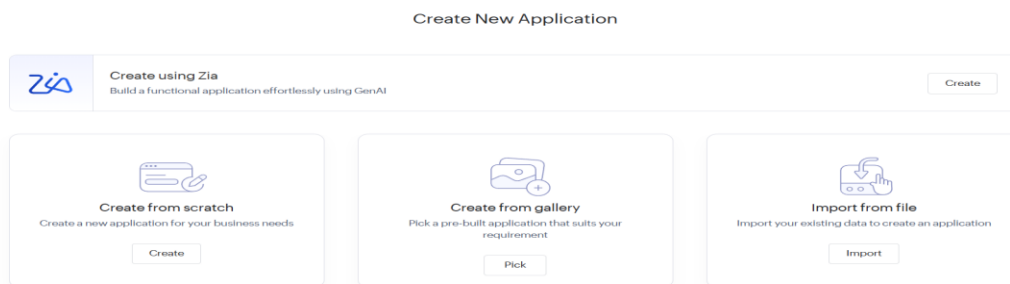
STEP1: GOTO ZOHOCOM



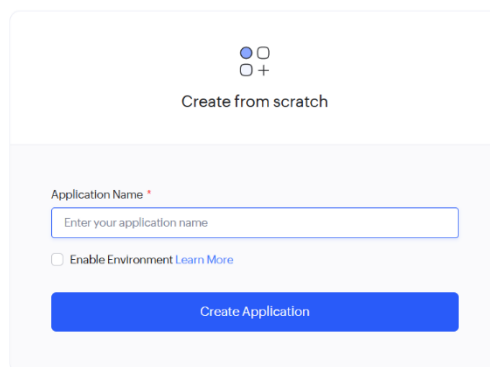
STEP 2: LOGINTOTHE ZOHOC.COM



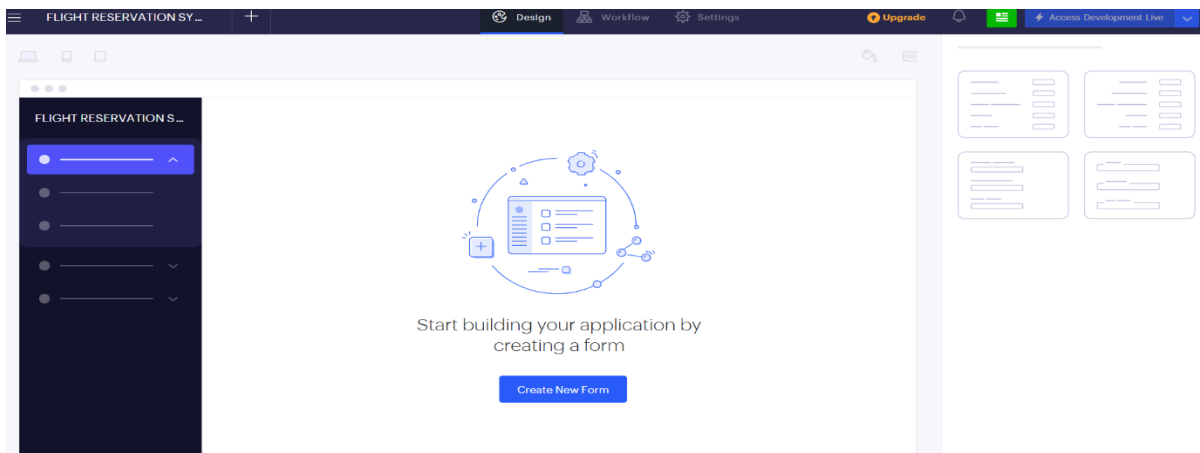
STEP 3: SELECT ONE APPLICATION



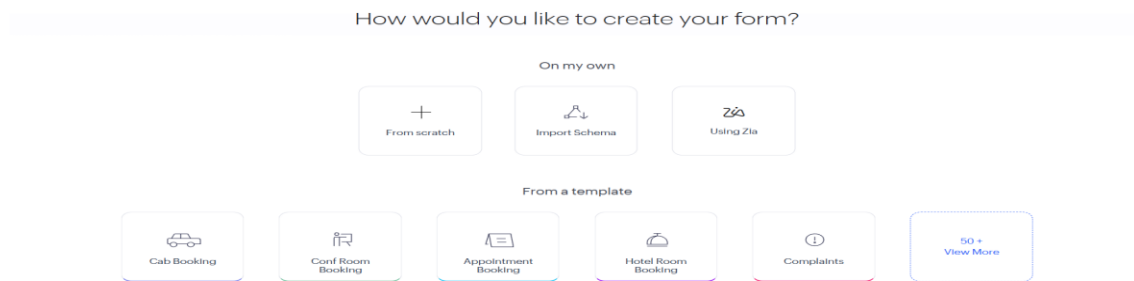
STEP 4: ENTER APPLICATION NAME



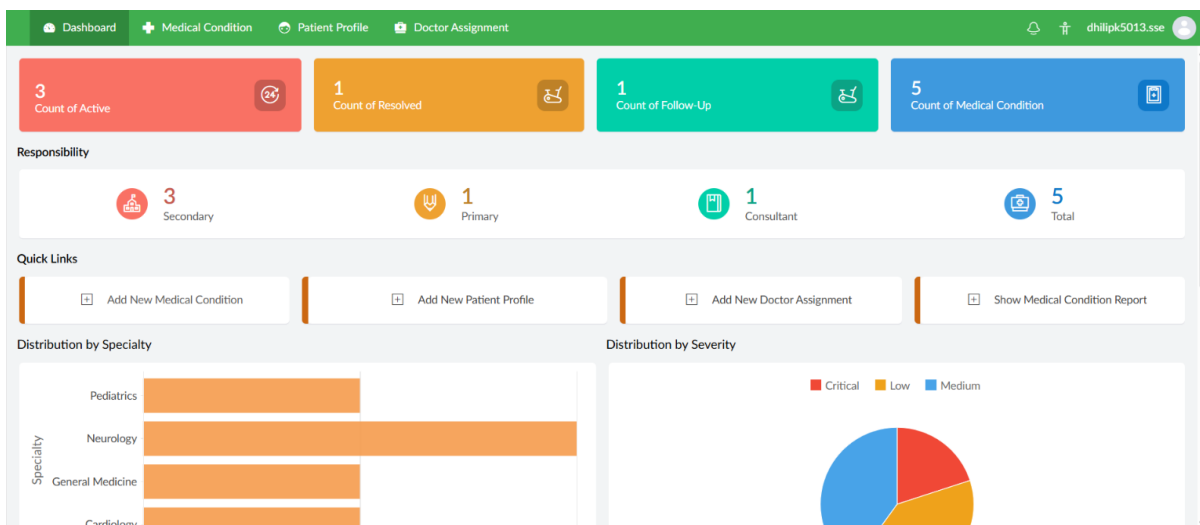
STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HAS BEEN CREATED.



The screenshot shows a web application with a green header bar containing navigation links: 'Dashboard', 'Medical Condition' (active), 'Patient Profile', and 'Doctor Assignment'. On the right of the header, there are icons for a bell, a user profile, and the text 'dhilipk5013.sse'. Below the header, the main content area is titled 'Medical Condition'. It contains a form with the following fields: 'Patient' (text input), 'Diagnosis Date' (calendar icon, placeholder 'dd-MMM-yyyy'), 'Condition Description' (text area), 'Treatment Plan' (text area), 'Status' (dropdown menu with '-Select-'), and 'Severity' (dropdown menu with '-Select-'). At the bottom of the form are two buttons: 'Submit' (green) and 'Reset' (grey).

3. Change the hardware compatibility of a VM (either by clone/create a new one) which is already created and configured, and run simple UNIX commands.

Aim

To change the hardware compatibility of an already created and configured Virtual Machine (VM) using VMware Workstation, and to run basic UNIX commands inside the VM.

Requirements

- VMware Workstation
- An already created Linux/UNIX Virtual Machine
- Existing VM configured and powered off

Procedure

Part 1: Change VM Hardware Configuration

1. Open VMware Workstation.
2. Select the already-created Linux VM from the left panel.
3. Power off the VM if it is running.
4. Right-click the VM and select Settings.
5. In the Hardware tab, you can modify the virtual hardware such as:
 - Memory (RAM)
 - Processors (CPU)
 - Hard Disk
 - Network Adapter
 - CD/DVD Drive
 - USB Controller
6. For example, select Memory and change the RAM from 2 GB to 4 GB.
7. Select Processors and change the number of processors/cores if required.
8. Click OK to save the changes.

9. Start the VM by clicking Power on this virtual machine.

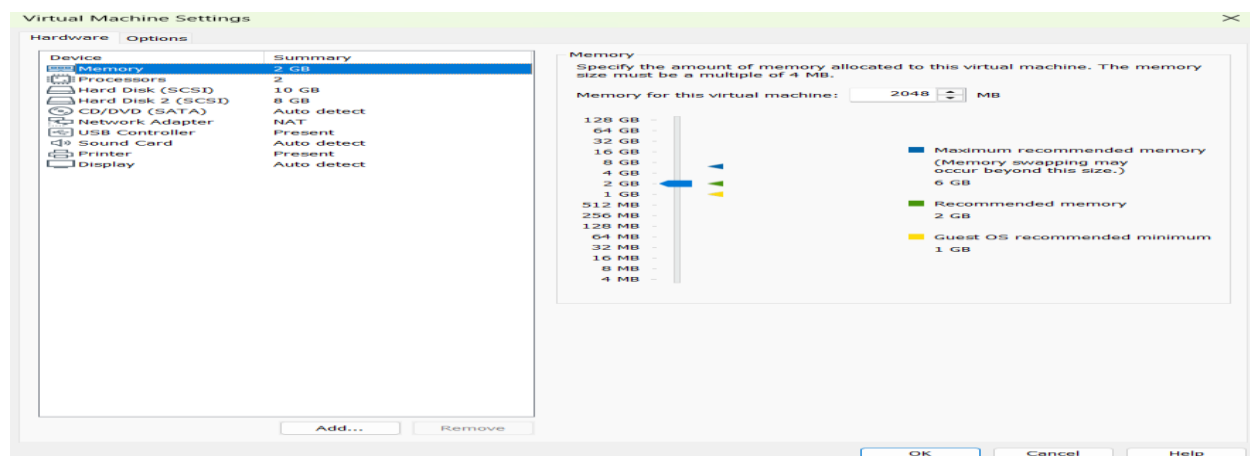
Part 2: Change VM Hardware Compatibility

If your VMware version provides the option:

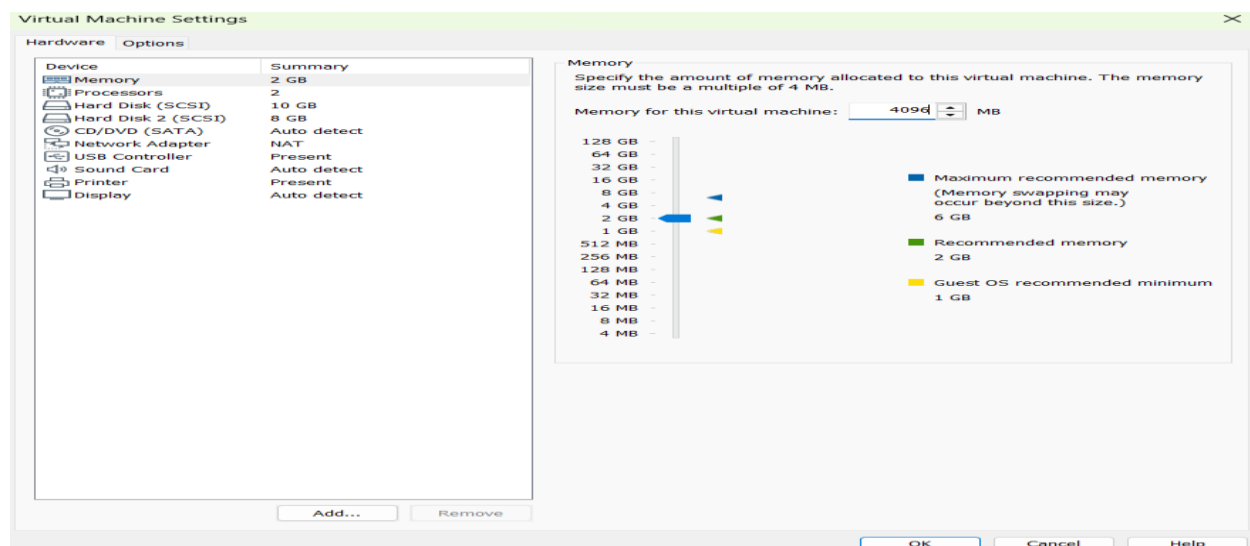
1. Make sure the VM is powered off.
2. Go to VM → Manage → Change Hardware Compatibility.
3. VMware displays the available hardware compatibility versions.
4. Select the required compatibility version.
5. Click Next.
6. Choose whether to create a new VM or update the existing VM, depending on the available option.
7. Follow the wizard and click Finish.
8. Power on the VM and verify that the operating system starts normally.

Implementation:

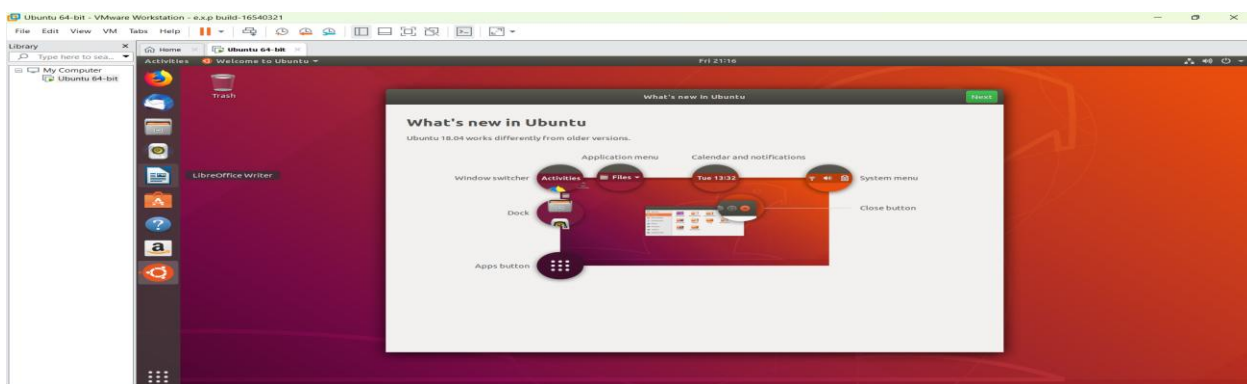
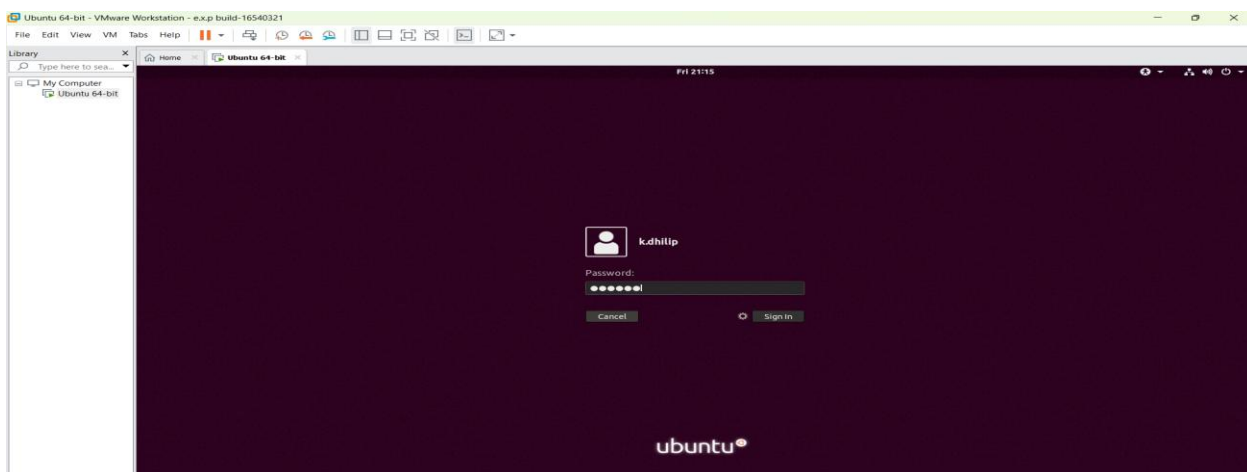
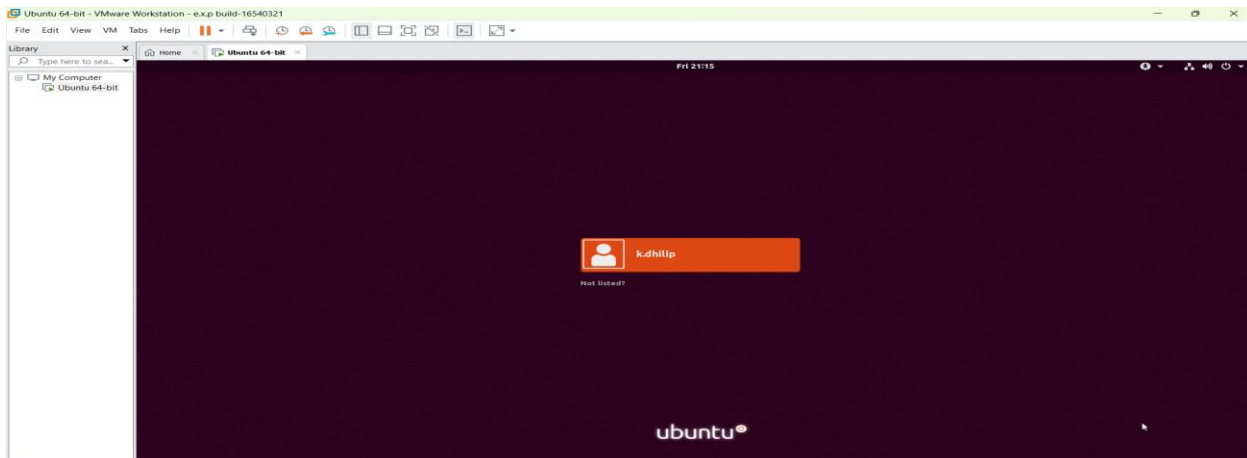
Step 1:



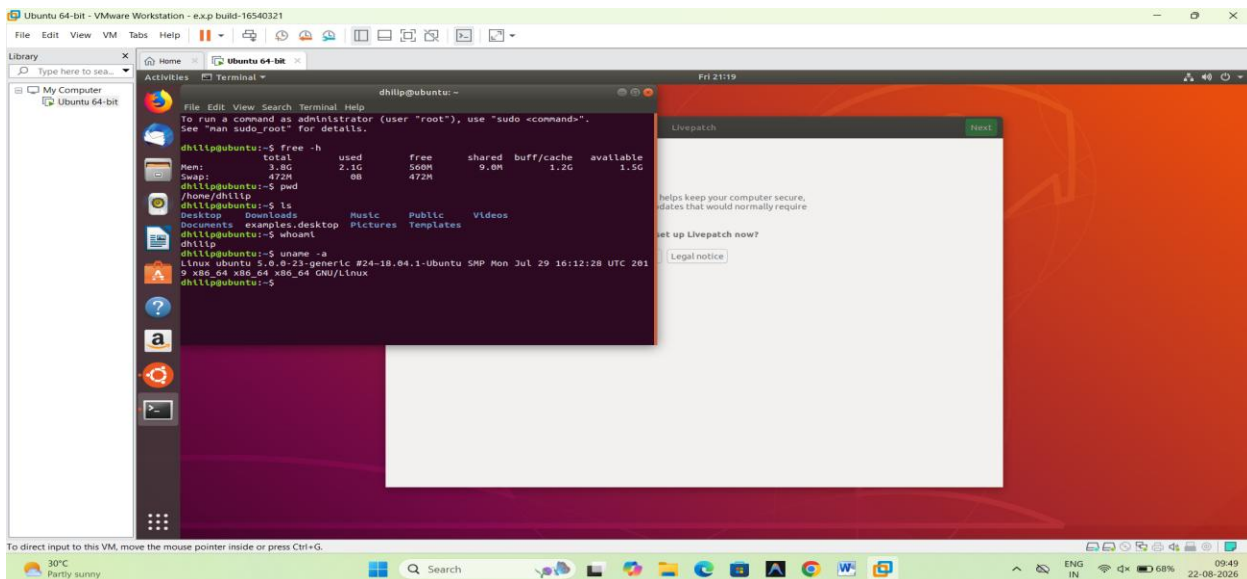
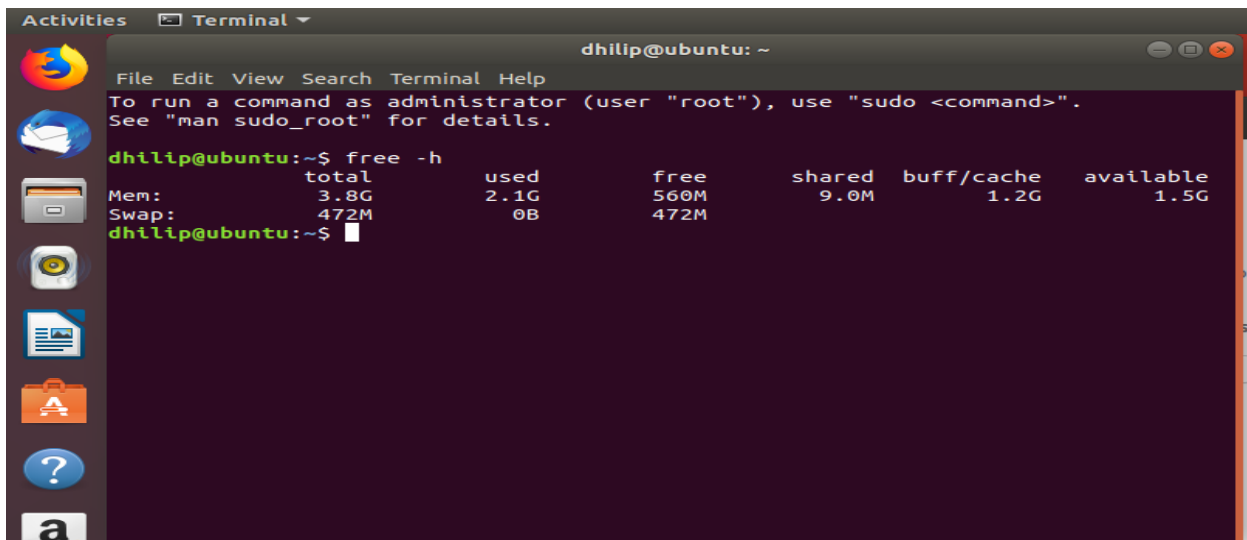
Step 2:

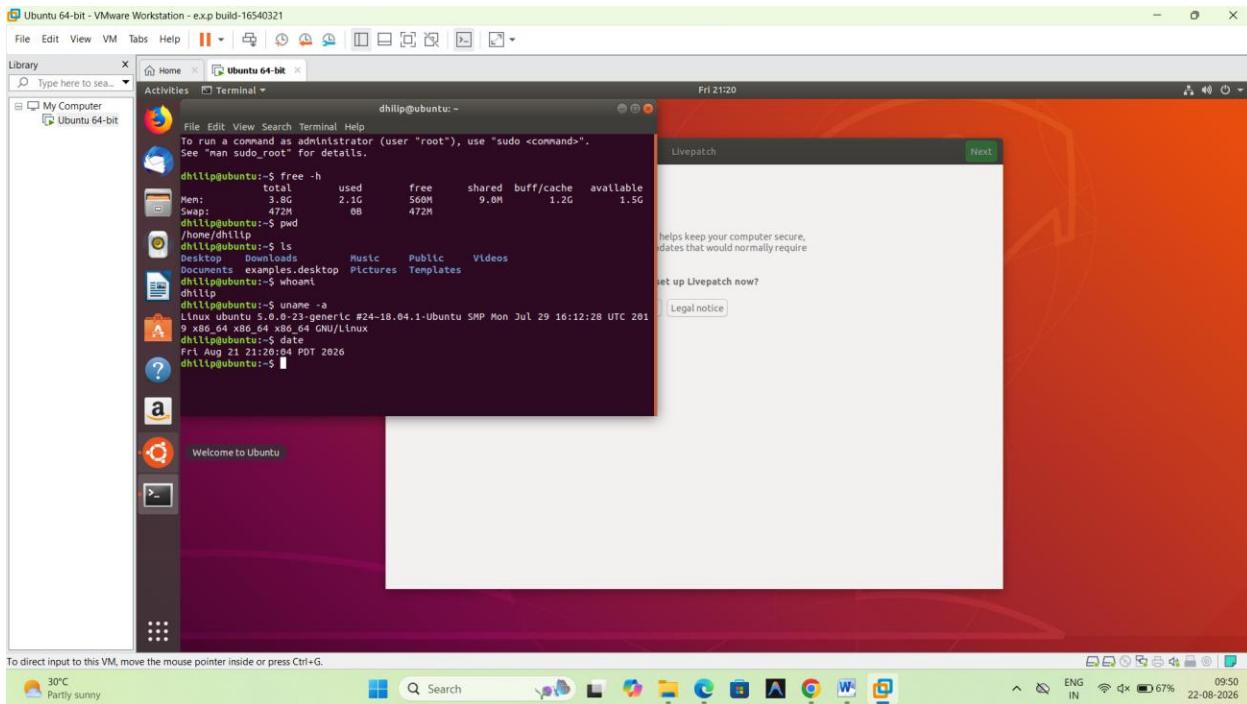


Step 3:



Step 4: Output:





4. Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

To create a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage disk using a Type 2 virtualization software (VMware Workstation).

Requirements:

- VMware Workstation
- Ubuntu/Linux ISO file
- Sufficient free space on the host computer

Procedure:

Step 1: Open VMware

1. Open VMware Workstation.
2. Click Create a New Virtual Machine.
3. Select Typical (Recommended).
4. Click Next.

Step 2: Select Operating System

1. Select Installer disc image file (ISO).
2. Click Browse.
3. Select your Ubuntu ISO file.
4. Click Next.

Step 3: Enter VM Information

Enter:

- Full Name: Your name
- Username: Your preferred username
- Password: Your preferred password

Click Next.

If VMware does not show this screen, you can install Ubuntu normally after creating the VM.

Step 4: Give VM Name and Location

For example:

- Virtual Machine Name: Ubuntu-15GB
- Location: Choose where you want to store the VM.

Click Next.

Step 5: Set Storage

When VMware asks for disk capacity:

- Maximum disk size: 15 GB
- Select Store virtual disk as a single file.

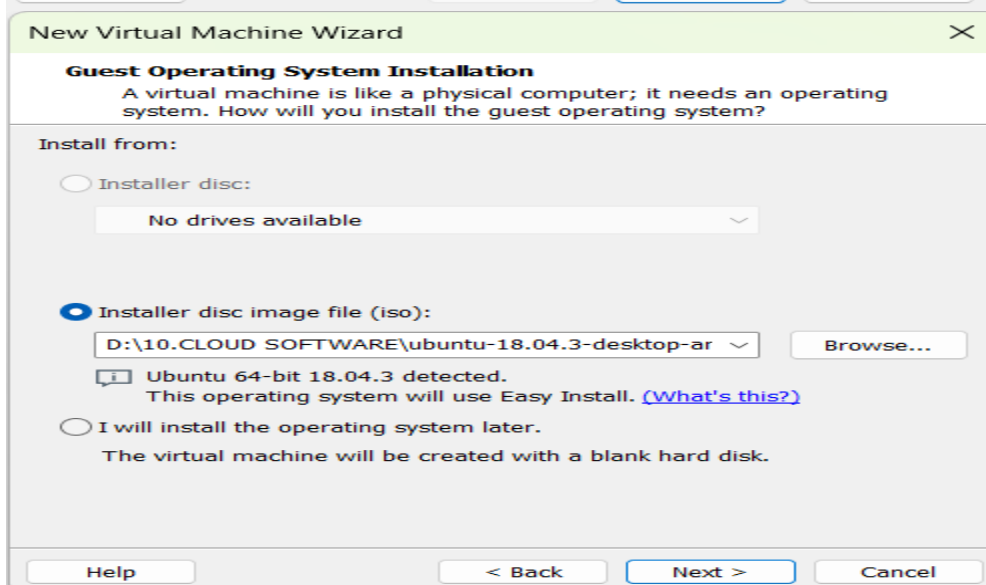
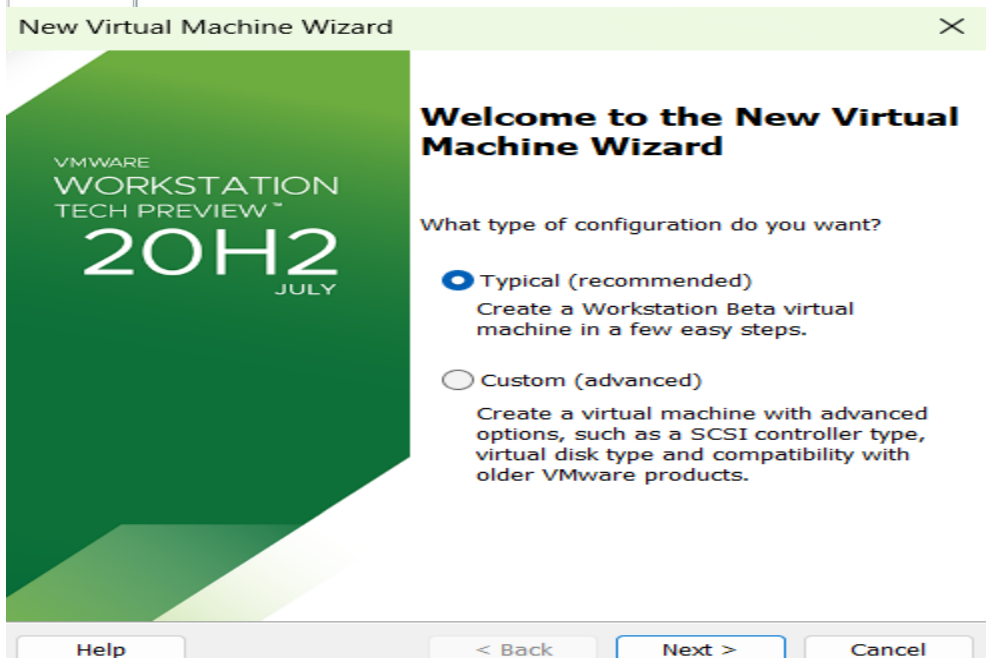
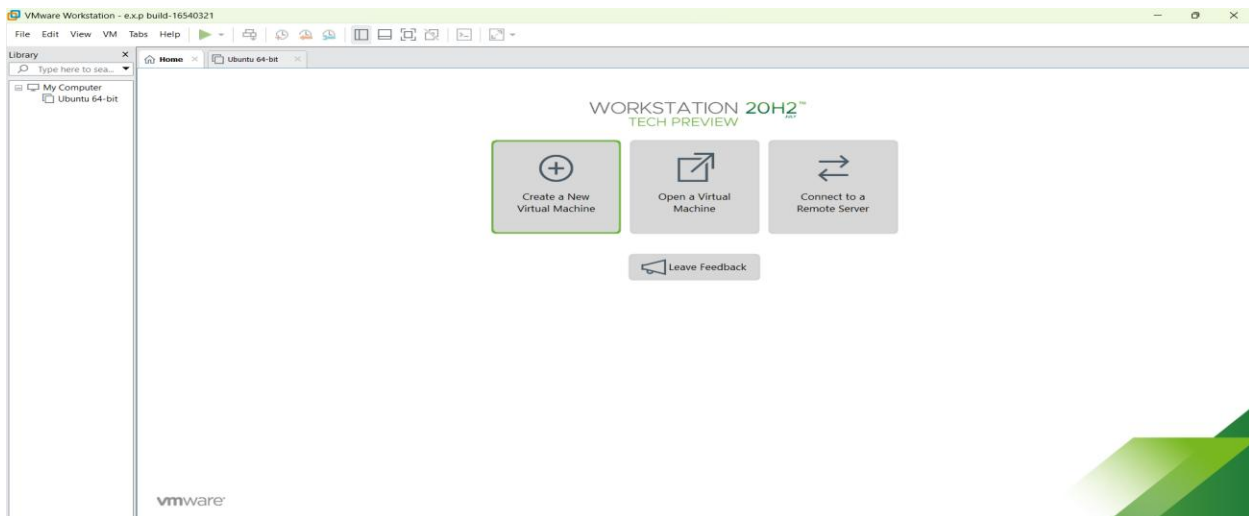
Click **Next**.

Step 6: Set CPU and RAM

Before finishing, click Customize Hardware.

Implementation:

Steps:



New Virtual Machine Wizard

Easy Install Information

This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

k.dhilip

User name:

dhilip

Password:

••••••

Confirm:

••••••

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu-15GB

Location:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-15GB

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

15

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

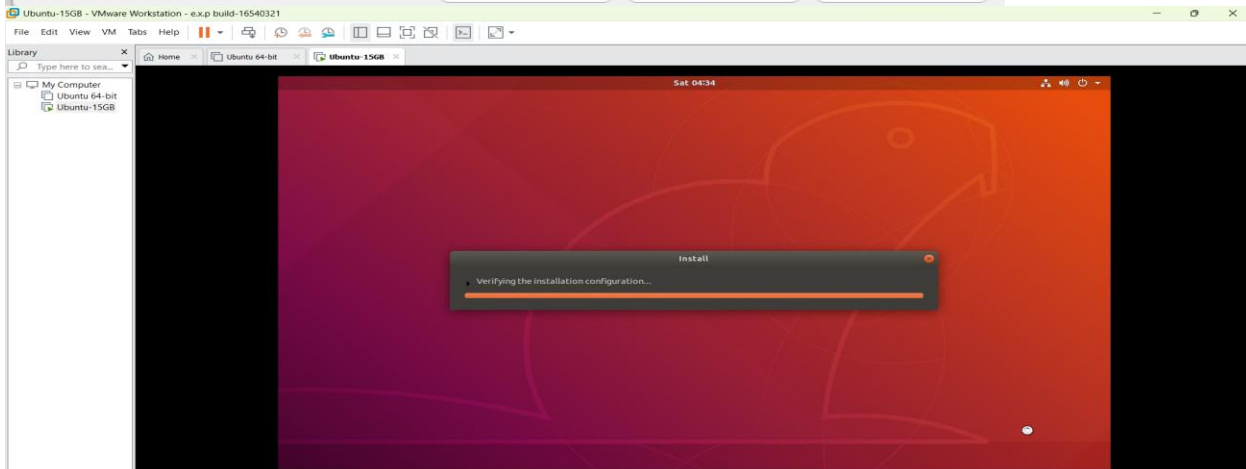
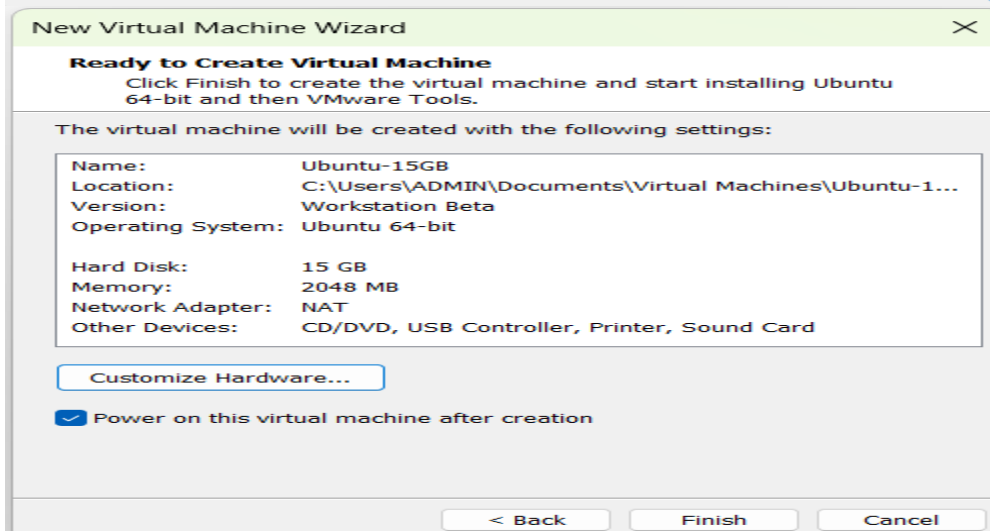
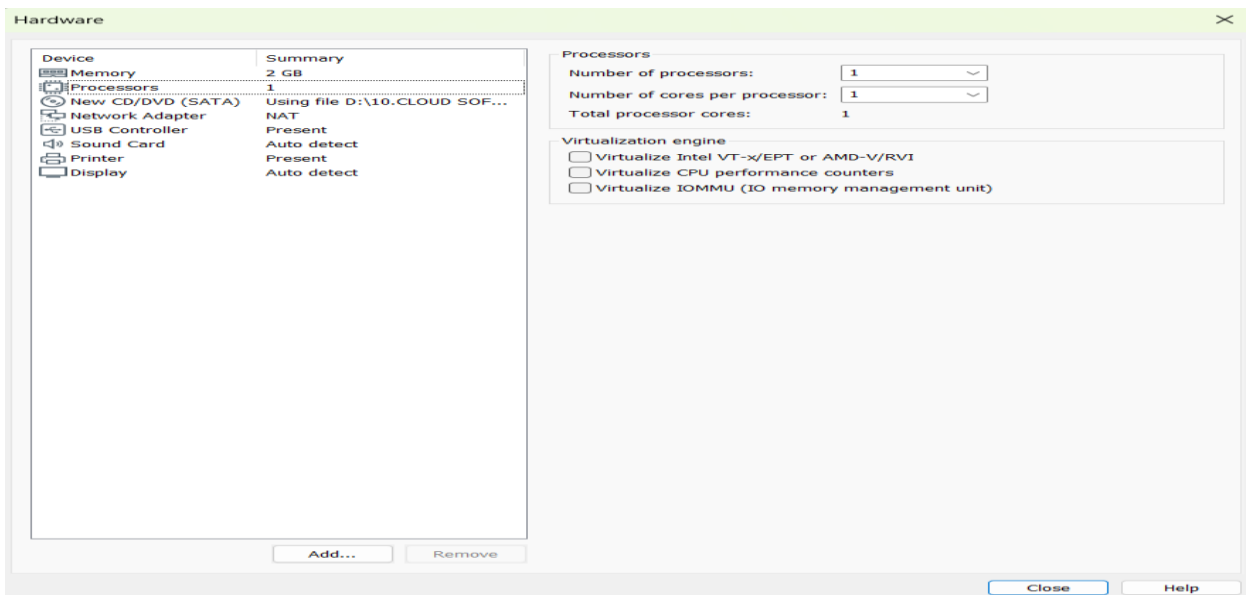
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel





k.dhilip

Not listed?



k.dhilip

Password:

Cancel



Sign In

